

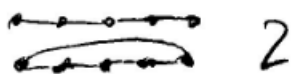
My favourite topics are Euler/Hamilton paths, Euler's formula for regions, graph planarity, and graph colouring $\rightarrow$ Kuratowski's theorem.

* 32. How many nonisomorphic simple connected graphs with five vertices are there

- a) with no vertex of degree more than two?
- b) with chromatic number equal to four?
- c) that are nonplanar?

[Redacted]

32 a) 5 vertices, non isomorphic simple connected graphs,



b) $\chi(G) = 4$, non isomorphic simple connected graph w/ 5 vertices

i.e. all graphs w/ 5 vertices s.t. $\deg(v) \leq 3 \forall v$.
 $\deg \leq 2$ has 2, above. $\deg \leq 3$ has

$$1 + 2C_1 + 3C_2 + 5C_3 + 2C_4 + 2C_5$$

$$+ 3C_1 + 2C_1 + 2C_6 - 2C_6$$



[Redacted]

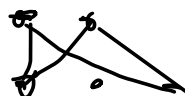
$$4[3C_1 + 3C_2] = 4[3 + 3]$$

$$= 24$$

$$3 + 1 + 1 + 2 + 4 + 3 + 2 + 3 + 2 + 1 = 12 + 4 = 16$$

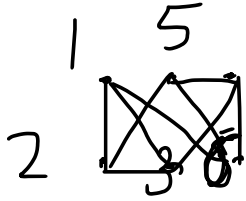
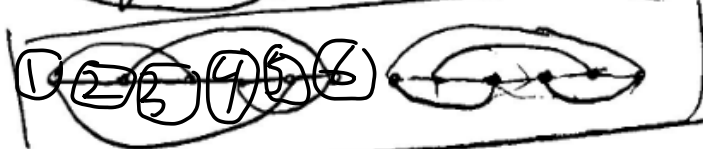
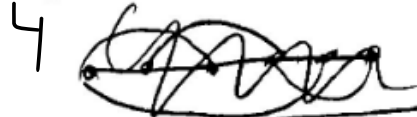


$$[Redacted] = 18$$

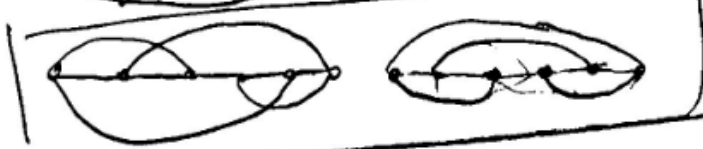
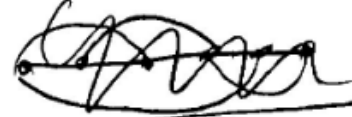


*47. Find the two nonisomorphic simple graphs with six vertices and nine edges that have optimal connectivity.

47)



47)



*53. a) Show that if the diameter of the simple graph G is at least four, then the diameter of its complement $\bar{G}$ is no more than two.

53a) if $d \geq 4$ of simple graph G , G has $d \leq 2$

$d \geq 4 \Rightarrow \geq 4$ edges $\Rightarrow \geq 5$ vertices
 $d=4$

if $d \geq 4 \Rightarrow \Delta(G)$ decreases monotonically.



Diameter $G_2 = 2 \leq 2$

diameter is max distance between 2 vertices.



$d=6$



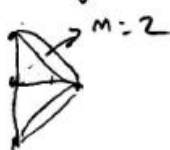
$d=3$

*54. Suppose that a multigraph has $2m$ vertices of odd degree. Show that any circuit that contains every edge of the graph must contain at least m edges more than once.

53b)

54) multigraph w/ $2m$ vertices, odd degree. ^{even}
 circuit w/ all edges
 Show every edge of a circuit contains at least

m edges more than once.



$m=2$

for every edge, there will be an extra input since odd degree. $\Rightarrow$ +1 repeated edge to leave. assume next edge traversal is not repeated.

$\frac{1}{2}$ repeated edges on minimum $\frac{1}{2} = \frac{2m}{2} = m$
 $\therefore$ repeated edges $\geq m$.

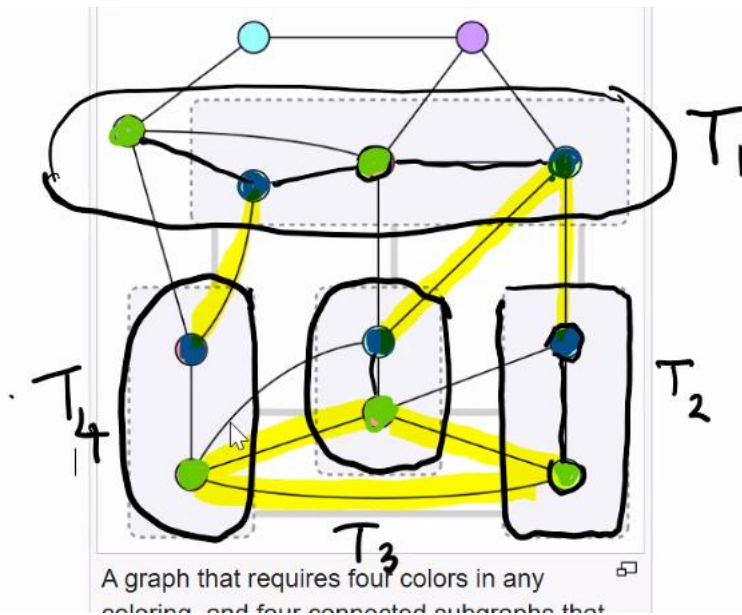
When assuming that the next node's edge traversal is not repeated, it means that the next node's edge is the i -th edge visited adjacent to the next node, where i is odd.

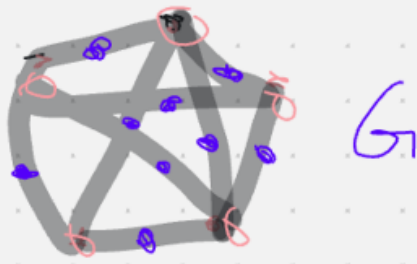
even degree nodes to consider.

**14. Work on the problem of determining whether the crossing number of $K_{7,7}$ is 77, 79, or 81. It is known that it equals one of these three values.

Computer problem.

A graph is a forest iff its flow semi group is aperiodic and a tree if its flow semi group is aperiodic and it contains a constant map.





G

but

$K_5 < G$
is a minor

$$\chi(G) = 2$$